

EXPERIMENT-06

Aim:

To demonstrate the performance of knn using wine dataset

Algorithm:

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from sklearn.datasets import load_wine
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```
# 1. Load Wine dataset
```

```
wine = load_wine()
```

```
X = wine.data
```

```
y = wine.target
```

```
# 2. Split dataset into training (80%) and testing (20%)
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.20, random_state=42, stratify=y  
)
```

```
# 3. Standardize features (important for KNN)
```

```
scaler = StandardScaler()
```

```
X_train_scaled = scaler.fit_transform(X_train)
```

```
X_test_scaled = scaler.transform(X_test)
```

```
# 4. Train KNN model with K = 5
```

```
knn = KNeighborsClassifier(n_neighbors=5)
```

```
knn.fit(X_train_scaled, y_train)
```

```
# 5. Predict test data
```

```
y_pred = knn.predict(X_test_scaled)
```

```
# 6. Evaluate performance
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("KNN Performance using Wine Dataset")
```

```
print("-----")
```

```
print("Value of K: 5")
```

```
print("Accuracy:", round(accuracy * 100, 2), "%")
```

```
print("\nClassification Report:")
```

```
print(classification_report(y_test, y_pred, target_names=wine.target_names))
```

```
# Confusion matrix
```

```
cm = confusion_matrix(y_test, y_pred)
```

```
plt.figure(figsize=(6, 4))
```

```
sns.heatmap(
```

```
    cm, annot=True, fmt="d", cmap="Blues",
```

```

        xticklabels=wine.target_names,
        yticklabels=wine.target_names
    )
plt.title("Confusion Matrix - KNN (K = 5)")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.show()

# 7. Check accuracy for several K values
k_values = range(1, 16)
accuracies = []

for k in k_values:
    model = KNeighborsClassifier(n_neighbors=k)
    model.fit(X_train_scaled, y_train)
    predictions = model.predict(X_test_scaled)
    accuracies.append(accuracy_score(y_test, predictions))

# Plot K value vs accuracy
plt.figure(figsize=(8, 4))
plt.plot(k_values, accuracies, marker="o", color="green")
plt.title("KNN Performance for Different K Values")
plt.xlabel("K value (Number of Neighbors)")
plt.ylabel("Testing Accuracy")
plt.xticks(k_values)
plt.grid(True)

```

```
plt.show()
```

```
# Display best K value
```

```
best_k = list(k_values)[accuracies.index(max(accuracies))]
```

```
print("Best K value:", best_k)
```

```
print("Best Testing Accuracy:", round(max(accuracies) * 100, 2), "%")
```

CODE:

```
 # KNN Performance Analysis using Wine Dataset

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# 1. Load Wine dataset
wine = load_wine()
X = wine.data
y = wine.target

# 2. Split dataset into training (80%) and testing (20%)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)
```

OUTPUT

